MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6205

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pentode, subminiature, sharp cutoff

Outline	---	3-1 (EIA)
Base	---	E8-10
Envelope	---	T3
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8
Element	---	g1	k sd	h	g3	a	h	g2	k sd

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ec3	Ehk	Rk
Unit:	V	Vdc	Vdc	Vdc	Vdc	v	Ohms
Maximum:	6.6	165	0,-55	155	22	200	---
Minimum:	6.0	---	---	---	---	---	---
TEST CONDITIONS:	6.3	100	0	100	0	0	150

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Rg1	Ik	Pp	Pg2	TE	Alt
Unit:	Meg	mAdc	W	W	°C	ft
Maximum:	1.1	16.5	1.1	0.55	+220	See note 1
Minimum:	---	---	---	---	---	---
TEST CONDITIONS:	---	---	---	---	---	---

GENERAL:

Qualification - Required

Reliable tube

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Method	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS BOGEY	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1256	Electrode current (1) (anode)		--- 0.4	--- II	Ib Ib	--- 5.5	7.5 ---	--- 9.5	mAdc mAdc
1256	Electrode current (2) (anode)	Ec1 = -9.0 Vdc; Rk = 0	0.4	II	Ib	---	---	50	µAdc
1256	Electrode current (screen)		0.4	II	Ic2	1.5	---	3.3	mAdc
1266	Total grid current	Rg1 = 1.0 Meg (see note 10)	0.4	II	Ic1	0	---	-0.3	µAdc
1301	Heater current		--- 0.4	--- II	If If	--- 140	150 ---	--- 160	mA mA
1306	Transconductance (1)		--- 0.4	--- II	Sm Sm	--- 4,200	5,000 ---	--- 5,800	µmhos µmhos
1336	Heater-cathode leakage		0.4	II	Ihk	---	---	5.0	µAdc
---	Suppressor	See note 3	0.4	II	---	---	---	---	---
1201	Short and discontinuity detection		0.4	II	---	---	---	---	---
	Quality conformance inspection, part 2								
1211	Insulation of electrodes		2.5	I	---	---	---	---	---
1031	Low-frequency vibration	Rp = 10,000 ohms; F = 40 Hz; 15 G	2.5	I	Ep	---	---	60	mVac
1246	Audio frequency noise	Esig = 70 mVac; Ec2 = 19 Vdc; Rg1 = 0.1 Meg; Rg2 = 1,000 ohms; Rp = 0.2 Meg (see note 4)	2.5	I	---	---	---	---	---
1266	Grid emission	Ef = 7.5 V; Ec1 = -9.0 Vdc; Rg1 = 1.0 Meg; Rk = 0 (see note 7)	2.5	I	Ic1	0	---	-0.5	µAdc
1306	Transconductance (2)	Ef = 5.7 V	2.5	I	ΔSm Ef	---	---	10	%
1311	Anode resistance		6.5	S3	rp	0.175	---	---	Meg
1331	Direct-interelectrode capacitance	0.405 in. dia shield	6.5	Code F	Cg1p Cin Cout	--- 3.5 2.9	--- --- ---	0.015 4.9 3.9	pF pF pF
1116	Lead fatigue		2.5	Code F	---	---	---	---	---
2126	Envelope strain		4.0	I	---	---	---	---	---
1041	Shock	450 G; Ehk = +100 Vdc (see note 8)	---	---	---	---	---	---	---
1031	Vibration fatigue	2.5 G; fixed frequency; F = 25 min, 60 max	6.5	See note 5	---	---	---	---	---

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Method	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS BOGEY	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued								
---	Post-shock and vibration fatigue-test end points:								
1031	Low-frequency vibration		---	---	Ep	---	---	200	mVac
1336	Heater-cathode leakage		---	---	Ihk	---	---	20	µAdc
1306	Transconductance (1), change of individual tubes		---	---	ΔSmt	---	---	20	%
1105	Permanence of marking		---	---	---	---	---	---	---

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC 1 ST SAMPLE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC COMBINED SAMPLES	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3									
1506	Heater-cycling life	Ef = 7.0 V; 1 min "on", 4 min "off"; Ehk = 140 Vac; Ec1 = Ec2 = Eb = 0	---	---	---	---	---	---	---	---
1516	Stability life	Ehk = +200 Vdc; Rg = 1.0 Meg; TA = room	---	---	---	---	---	---	---	---
---	Stability life-test end point:									
1306	Transconductance (1), change of individual tubes		---	---	---	---	ΔSmt	---	10	%
1501	Intermittent life (room temperature)	Stability life test, or equivalent conditions; TA = room (see notes 6 and 9)	---	---	---	---	---	---	---	---
---	Intermittent life-test end point: (room temperature) (500 hours)									
1211	Insulation of electrodes		---	---	1	3	R	50	---	Meg
---	Intermittent life-test end point: (room temperature) (1,000 hours)									
1211	Insulation of electrodes		---	---	1	3	R	25	---	Meg
1501	Intermittent life (high temperature)	Stability life-test conditions: TE = +220°C (min) (see note 2)	---	---	---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC		SYMBOL	LIMITS		UNIT
					1ST SAMPLE	COMBINED SAMPLES		MIN	MAX	
	Quality conformance inspection, part 3 -Continued									
---	Intermittent life-test end points: (high temperature) (500 hours)									
---	Inoperatives		---	---	1	3	---	---	---	---
1266	Total grid current	See note 10	---	---	1	3	Icl	0	-0.8	µAdc
1301	Heater current		---	---	1	3	If	138	164	mA
1306	Transconductance (1) change of individual tubes		---	---	1	3	ΔSmt	---	20	%
1306	Transconductance (2)		---	---	1	3	ΔSmEf	---	15	%
1336	Heater-cathode leakage		---	---	1	3	Ihk	---	10	µAdc
1211	Insulation of electrodes		---	---	1	3	R	50	---	Meg
1306	Transconductance (1),average change		---	---	---	---	Avg ΔSmt	---	15	%
---	Total defectives		---	---	3	6	---	---	---	---
---	Intermittent life-test end points: (high temperature) (1,000 hours)									
---	Inoperatives		---	---	1	3	---	---	---	---
1266	Total grid current	See note 10	---	---	1	3	Icl	0	-0.8	µAdc
1301	Heater current		---	---	1	3	If	138	164	mA
1306	Transconductance (1), change of individual tubes		---	---	1	3	ΔSmt	---	25	%
1306	Transconductance (2)		---	---	1	3	ΔSmEf	---	20	%
1336	Heater-cathode leakage		---	---	1	3	Ihk	---	10	µAdc
1211	Insulation of electrodes		---	---	1	3	R	25	---	Meg
---	Total defectives		---	---	4	8	---	---	---	---

NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage.
2. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a tube having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.
3. The tube shall be subjected to the following test, or equivalent. Reject for open suppressor if anode current does not decrease by a minimum of 10 percent when Ec3 is changed from 0 to -100 Vdc.
4. The rejection level shall be set at the VU meter reading obtained during calibration.
5. This test shall be conducted on the initial lot and thereafter on a lot approximately every 6 months. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. When one lot has passed, the 6-month rule shall apply. MIL-STD-105, sample size code letter E, shall apply.
6. Eligibility for early release at 500 hours is established if no lot failure due to this life test or the 1,000-hour intermittent (high temperature) life test has occurred in the preceding three consecutive lots and the lot under consideration is acceptable at 500 hours. One lot failing the 1,000-hour intermittent (high temperature) life test or the 1,000-hour intermittent (room temperature) life test, shall result in loss of eligibility for the early release procedure. All intermittent (room temperature) life-test samples shall be operated to 1,000 hours.

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7. Prior to this test, tubes shall be preheated 5 minutes at the conditions specified below. The 3-minute test is not permitted. Test at specified conditions within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid-emission test.

Ef	Ec1	Ec2	Ec3	Eb	Rk	Rg1
V	Vdc	Vdc	Vdc	Vdc	Ohms	Meg
7.5	0	100	0	100	150	1.0

8. A grid resistor of 0.1 megohm shall be added; however, this resistor shall not be used when a thyratron-type short indicator is used.

9. The life-test sample shall consist of 20 tubes per lot and not more than 1 tube failure shall be permitted. In the event of rejection of the first sample, due to failure of more than 1 tube, a second sample of 40 tubes shall be selected from the lot. Acceptance shall then be based on the combined first and second samples. The total tube failures from the combined first and second samples shall not exceed three.

10. This test to be performed at the conclusion of the holding period.

11. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

User activities:

Navy - AS, OS, MC, CG
Air Force - 11, 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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